

Planets and Orbits 5E

Is the exoplanet like Earth in terms of its distance from its star? What are the factors that allow an exoplanet to maintain a stable temperature?

Performance Expectations
HS-ESS1-4

Investigative Phenomenon
Comet Borrelly has water that is frozen most of the time, but every several years it shoots out a jet of vaporized water and dust.

Time
7-9 days

In this 5E instructional sequence, students are investigating the following questions that surfaced during the Driving Question Board launch - *Does the planet have the right temperature to sustain life? Does the Planet have liquid water?* This leads to questions about whether the average distance between a star and an exoplanet allows for the possibility of liquid water's existence throughout a period of revolution. Through graphical and algebraic analysis and interpretation of data, students investigate the orbits of planets and objects in our solar system. Through this exploration they determine relationships among orbital variables that are represented by Kepler's Laws. Students then apply what they have learned about orbits and habitable zones to exoplanet data and argue from evidence about which planet is most likely to maintain a temperature that allows for the existence of liquid water throughout a revolution around its star.

Engage	Why does water on Comet Borrelly change from ice to vapor every 6 years?	Students develop an initial solar system model to show how characteristics of orbits might explain the phenomenon of phase change during a comet's revolution around the Sun and generate questions about what data they would like to investigate.
Explore 1	Identifying patterns in solar system data to test their initial models	Students are provided with orbital data from our solar system and graph it in order to identify evidence of patterns and test their explanatory models for phase change and stability of water on Comet Borrelly and other objects that orbit the Sun.
Explain 1	Revising solar system models	Students use the evidence of patterns identified in graphs they created to revise their solar system models that explain water's phase change and stability on objects orbiting the Sun. They use their models to identify important features of the motions of orbiting objects that would allow a planet to maintain liquid water.
Explore 2	Developing a mathematical model for solar system orbits	Students use algebraic thinking to examine graphs from the Explore 1 in order to create a mathematical representation that allows them to identify the exact pattern in the relationship between orbital period and average distance between a planet/object and its star .
Explain 2	Using a mathematical representation to determine the orbit of exoplanets	Students use a mathematical equation representing the relationship between orbital semi major axis and orbital period (Kepler's Third Law) to make predictions about exoplanets' orbital semi major axis based on each exoplanet's orbital period .
Elaborate	Constructing orbits of exoplanets to determine whether liquid water can exist on each planet	Students use a mathematical model (Kepler's First Law) to describe the relationship between orbital eccentricity, major axis, and distance between foci of a planet/object revolving around its host star . Students then use the mathematical model and algebraic thinking to construct exoplanet orbits, allowing them to make predictions about which exoplanets stay within the habitable zone of their stars throughout an entire orbit.
Evaluate	Developing or revising models to include new ideas about orbits	Students develop new models or revise existing ones to represent the idea that stable temperatures and the maintenance of liquid water result from features of the motions of orbiting objects .

Engage

Why does water on Comet Borrelly change from ice to vapor every 6 years?

Students **develop** an initial solar system **model** to show how characteristics of **orbits** might explain the phenomenon of phase change during a comet's revolution around the Sun and **generate questions** about what data they would like to investigate.

Preparation

Student Grouping

- ☐ Independent
- ☐ Small group

Routines

- ☐ Rumors

Literacy Strategies

- ☐ Text Annotation

Materials

Handouts

- ☐ Which parts of the solar system have liquid water?

Lab Supplies

None

Other Resources

- ☐ Sticky notes
- ☐ [Comet Borrelly Flyby](#)

Launch

1. Remind students that, during the Driving Question Board launch, one category of questions that emerged was related to the temperature and the existence of liquid water on the exoplanets (for example, *Does the planet have the right temperature to sustain life? Does the Planet have liquid water?*).
2. Point out to students that the exoplanet data does not tell them whether water exists on the planets because so far our technology only allows us to detect the existence of water within our own solar system. Ask students: "How can we use what we know about water in our solar system to make claims about the potential for liquid water on exoplanets?" Students may propose to investigate the places where liquid water exists in our solar system and why it exists in these places.

3. Introduce the handout *Which parts of the solar system have liquid water?*, which provides some examples of planets and other objects in our solar system that have water in its different phases (ice, liquid, and vapor).
4. Show students a video of Comet Borrelly, to get them thinking about how the comet is different from other objects in the solar system.

Implementation Tip



The performance task for this unit asks students to develop a model of what makes Earth a place where life can exist, as a way to make claims about the habitability of exoplanets. If students don't suggest to investigate water in our solar system in order to make claims about water on exoplanets, refer back to the performance task prompt as a way to support them in arriving at the idea that we should study water in our solar system in order to make claims about water in other solar systems.

Surfacing Student Ideas

1. Have students read the *Water in Our solar system* table independently and use the **text annotation** strategy to identify 3-5 details they think are important to telling the story of the phenomenon of water in our solar system.

Differentiation Point



The *Water in Our solar system* data table may be overwhelming for some students at first. Students who need more support can start with a smaller set of data (just Mercury, Earth, Jupiter, and Comet Borelly, for example). The teacher can also guide the text annotation work by asking questions like the following:

- Which planets are closest to and farthest from the Sun?
- What does "average distance" mean?
- What are all the forms of water that are named in the middle column?

2. Support students' thinking about the table by asking the following questions:

Access for All Learners



While all students have some background knowledge on the topic of water existing in and changing into different phases, it's likely that most of them have never thought about this within the context of our solar system. Asking students to think about a similar phenomenon they are familiar with will help them connect to and find relevance in ideas about energy they already know. This will help all students

- a. Why do you think the large majority of water on Earth exists as liquid, but most if not all water on other solar system objects in the table exists as ice or vapor?
- b. What might explain the fact that Comet Borrelly has water that is frozen most of the time, but every several years shoots out a jet of vaporized water and dust?

3. Have students work in small groups to share the important details they underlined, then write a description of the phenomenon.

Implementation Tip



Make sure that students understand what “average distance” means in this table, without giving away the point that they will be figuring out in this 5E lesson, which is that comets have a low average distance from the Sun because they spend part of their orbit very far away, and part of it very close by.

Describe the phenomenon

Talk with your group and tell the story of the phenomenon captured in the video and the *Water in the solar system* table.

The large majority of water on Earth exists as liquid, but most if not all water on other solar system objects in the table exists as ice or vapor.

Comet Borrelly has water that is frozen most of the time, but every several years shoots out a jet of vaporized water and dust.

4. Have students turn to a partner and discuss the following prompt about related phenomena: *What other phenomena are you familiar with that are associated with water in different places existing in different phases?*
5. Ask students to independently develop an initial model that explains the phenomenon.
6. Have each student write a response to the question: *What might explain the fact that Comet Borrelly has water that is frozen most of the time, but every several years shoots out a jet of vaporized water and dust?* on a sticky note.
7. Use the Group Learning Routine **Rumors** to have all students share their initial ideas with each other.

generate initial ideas about why water exists in different phases on different objects in our solar system and why it changes phase on Comet Borrelly.

Routine



Rumors is a routine designed to surface all students’ initial ideas in a low stakes manner. After having an opportunity to independently brainstorm, students identify the idea they are most confident about and share it with several classmates. Students listen for trends in their thinking and at the end of the routine share those trends with the rest of the class including the teacher. Please read the Earth & Space Science Course Guide for detailed steps about this routine.

Access for All Learners



All students have some background knowledge about why water exists in different phases, and this Engage phase is designed to support students in applying their existing intuitions to making sense of solar system data. Incomplete ideas are evidence of sensemaking, and there will be

8. After students have shared their ideas through Rumors, categorize student ideas in order to address during the instructional sequence. Students are likely to have a range of ideas.

plenty of opportunities for students to develop an accurate explanation of Comet Borrelly's behavior.

Look & Listen For



Students have background knowledge that can be used to drive the investigation. Listen for the following ideas related to energy and its conservation that students grappled with in middle school:

- Somehow the Sun is transferring more electromagnetic energy to Comet Borrelly every several years, increasing the average kinetic energy (temperature) of water molecules enough for them to vaporize (MS-PS1.A, MS-PS3.A, MS-PS4.B)
 - *Students will be able to confirm this idea when they analyze and interpret the solar system Orbital Data table in the Explore 1 phase.*
- Maybe the electromagnetic waves from the Sun hit water on the comet with more intensity every several years because it's its tilt with respect to the earth changes, like seasons on Earth (MS.ESS1.B, MS-PS4.B)
 - *When students analyze and interpret the solar system Orbital Data table in the Explore 1 phase, they will find no evidence that comet Borrelly experiences seasons because of a tilted axis like Earth. They will instead find that the distance at which the comet orbits around the sun decreases significantly during its orbit, which could easily account for the phase change of water every six years.*

Classroom Supports



Create a poster or space on a whiteboard for categories of student ideas that have come up. Use the title *Water in Our solar system*. Use this tracking tool to return to ideas in the unit.

9. Tell students that the class will need to investigate further in order to test their ideas about why Comet Borrelly has water that is frozen most of the time, but every several years shoots out a jet of vaporized water and dust.
10. Ask students to independently generate at least 3 questions they want to investigate further in order to explain the Comet Borrelly phenomenon. Have them turn to a partner and take turns sharing questions, then decide which two questions they want to share with the rest of the class.

Look & Listen For



Students may surface these questions that are relevant to this phenomenon and can be used to transition to analyzing solar system orbital data in the Explore phase:

- At what distance does Comet Borrelly orbit our Sun during different points of its orbit?
- What is Comet Borrelly's orbit like? What are its characteristics?
- How is Comet Borrelly's orbit different from orbits of other objects in the solar system?
- What patterns do we see in the orbits of objects with water in solid phase? Gas? Liquid?

Differentiation Point



If students are able to demonstrate a complete understanding of why water exists in different phases on different celestial objects in our solar system and why water changes phases on Comet Borrelly, consider having them skip the Explore 1 and Explain 1 phases, which support the development of those explanations. If you have students skip the Explore 1 and Explain 1 phases, graphing of orbital variables still needs to be completed prior to the Explore 2 phase.

Explore 1

Identifying patterns in solar system data to test their initial models

Students are provided with **orbital data** from our solar system and **graph it in order to identify evidence of patterns** and test their **explanatory models** for phase change and stability of water on Comet Borrelly and other objects that orbit the Sun.

Preparation

Student Grouping

☐ Groups of 2-3 Students

Routines

☐ Domino Discover

Literacy Strategies

None

Materials

Handouts

- ☐ Orbital Data Investigation
- ☐ Making Sense of the Orbital Data Investigation
- ☐ Orbital Data Investigation Rubric

Lab Supplies

None

Other Resources

- ☐ Computers with Excel or Google Sheets
- ☐ [Excel Spreadsheet for Planet Data Analysis Graphing](#)
- ☐ *Orbital Data Investigation Student Work*

Launch

1. Remind students that, at the end of the Engage phase, they surfaced the idea that they need to investigate the distance Comet Borrelly is from our Sun and the characteristics of Comet Borrelly's orbit.
2. Tell them that they will be looking for evidence of patterns in orbital data from our solar system in order to make sense of the phase changes of water on Comet Borrelly.

Analyzing data about objects in the solar system

1. Provide each student with the handout *Orbital Data Investigation*
2. Ask students what they think they should look for in the data in order to better understand the water in the solar system phenomenon. Listen for students to say they want to look for evidence of patterns and leverage that to move to step 3.
3. Ask students to turn and talk to a partner and discuss the following question:
 - What have you done in the past to represent data in a way that makes it easier to identify evidence of patterns?

Students will probably offer the ideas of “creating a graph” and “looking for trends.” If they don’t prompt them to think about what they did with the Star Circle data in order to identify evidence of patterns. Once students have said they want to graph the orbital data use that transition to step 4.

4. For New York State teachers, take this opportunity to have students turn to the *Solar System Objects Data Table* on page 2 of the New York State Earth & Space Science Reference Tables. Have students look for similarities and differences in the type of information they can gain from each data source. Explain to students that AUs and million kms are different ways to measure distances, but that they represent the same measurement. Point out to students that the table on their handouts lists Europe instead of Jupiter and Mimas instead of Saturn, but that the distance of the moons from the sun is about the same as the distance of the planets from the sun.
5. Provide students with computers that have a spreadsheet application like Excel or Google Sheets.
6. Support students in making decisions about what data they would like to graph by encouraging them to look for possible patterns/relationships they would like to confirm or see more clearly by graphing the data.
7. As students complete their graphs using the [Excel Spreadsheet for Planet Data Analysis Graphing](#) and make interpretations, confer with them to support their sensemaking.

Integrating Three Dimensions



Students have had multiple opportunities to consider **CCC #1 - Patterns** so far in this unit. In this phase, the idea that graphs and charts are useful for identifying patterns in data is an important middle school element of the CCC. Prompt students to think about how they can represent the data in order to more easily identify evidence of patterns. This will make it easier to foreground this element of Patterns at the end of this phase and build toward an important high school element of patterns during the Explore 2 and Explain 2 phases.

Implementation Tip



If students have never used Sheets or Excel to create graphs, it may be necessary to take time out for a tutorial at this point. It is important that students create and analyze graphs independently, as this is how they develop the practice of modeling.

Conferring Prompts



Confer with students as they work in collaborative groups to complete the graphs.

Suggested conferring questions:

- What data do you think is most relevant to explaining why water exists in different phases on different celestial objects and why water on Comet Borrelly changes phases during its orbit?
- What patterns do you see in the solar system orbital data that could help you explain the Comet Borrelly investigative phenomenon?

Whole-Class Investigation Summary

1. After students have had an opportunity to analyze and interpret the data, have them independently complete the handout *Making Sense of the Orbital Data Investigation*.

Conferring Prompts



TIP: Confer with students as they work in collaborative groups to analyze data and complete the See-Think-Wonder chart.

Suggested conferring questions (these should push students' thinking around establishing relationships, observing patterns, identifying variables, and questioning events):

- What patterns do you see in the relationship between the phase of water on solar system objects and their average distance from the Sun?
- What pattern do you see in the minimum and maximum distance from the Sun?
- What patterns do you see in the relationship between an object's average distance from the Sun and orbital period?
- Do all objects in the solar system fit that pattern?
- Which patterns or exceptions to patterns in the data that could help us explain the Comet Borrelly investigative phenomenon?

Routine



The **Domino Discover** group learning routine is an opportunity to surface students' thinking to the whole class and the teacher. It allows students to learn from each other and for the teacher to assess whether the class is ready to move to the next phase of instruction. Refer to the Earth & Space Science Course Guide for support with this routine.

2. Elicit student ideas through the group learning routine, **Domino Discover**.

Look & Listen For



There are several patterns students may see in the solar system orbital data that could help you explain the Comet Borrelly investigative phenomenon. Some possible responses that show students are making sense of the include these responses to the See-Think-Wonder

See

- The planets and objects that have water in solid ice form are further from our Sun than Earth, where water is mostly liquid
- The planets and objects that have water in vapor/gas form are closer to our Sun than Earth
- The distance at which Comet Borrelly and Pluto revolve around the Sun changes a lot compared to other planets and objects in the solar system
- As the average distance at which an object orbits around the Sun increases, so does the orbital period

Think

- I think this is because the further away you get from our sun, the colder it is, so water is frozen
- I think this is because the closer you get to the sun, the hotter it is, so water is in vapor form
- Earth is at just the right temperature for most of our water to exist in liquid form
- I think this might be the reason Comet Borrelly has water that is frozen most of the time, but every several years shoots out a jet of vaporized water and dust
- I think this is because if a celestial object orbits further away, its orbit is longer so it takes longer

Wonder

- Do the planets that have water in mostly ice and vapor form have any liquid water? Did they ever?
- Does comet Borrelly ever have water in liquid form?
- Do the exoplanets we're investigating orbit around their stars at a distance that allows water to be in liquid form? Do they stay at that distance throughout their entire orbit around their star?

Implementation Tip



At this point it's ok if students only name the distance a planet travels as a factor that contributes to the relationship between average distance from the Sun and orbital period. An understanding of the role gravity plays in determining this relationship will be developed in Unit 2 of this course.

3. If students don't surface one or more of the observations above, display student graphs and use the suggested conferring questions from this Explore 1 phase to have students surface those ideas. Then ask students to share their thoughts about whether any of the newly surfaced patterns are relevant to our questions regarding water existing in different phases on different objects in our solar system.

Differentiation Point

- Some students may have difficulty identifying relevant patterns in the solar system orbital data.
- Differentiate for these students by including sentence starters in the 'See' column of the
- See-Think-Wonder organizer that support them in identifying those patterns.

4. Provide students with *Orbital Data Investigation Rubric*. Ask students to use the investigation rubric to self and peer assess their progress on engaging with the investigation individually and as a group.

Explain 1

Revising solar system models

Students use the **evidence of patterns identified in graphs** they created to **revise their solar system models** that explain water's phase change and stability on objects orbiting the Sun. They use their models to identify important **features of the motions of orbiting objects** that would allow a planet to maintain liquid water.

Preparation

Student Grouping

☐ Groups of 2-3 Students

Routines

None

Literacy Strategies

None

Materials

Handouts

- ☐ Revising Solar System Models
- ☐ Summary Task

Lab Supplies

None

Other Resources

Launch

1. Remind students that we are trying to figure out why water exists in different phases on different celestial objects and why water on Comet Borrelly changes phases during its orbit.
2. Tell students that they will now have an opportunity to use evidence of patterns in orbital data to revise their initial solar system models from the Engage phase.

Revising Models

1. Have students work in small groups to revise their initial models.
2. Confer with groups as they work to understand how they are thinking about the phenomenon, and as a way to get a sense of the range of student ideas in the classroom.

Look & Listen For



As students are revising their models, look and listen for these ideas, which will be important to surface in the Class Consensus Discussion:

- Celestial objects on which water exists mostly as vapor orbit closer to the Sun than Earth.
- Celestial objects on which water exists mostly as ice orbit further from the Sun than Earth.
- Comet Borrelly and Pluto both have orbits that are more elliptical than the other orbits.
- Annotations that articulate how the orbits explain the most abundant phase of water on celestial objects.

Class Consensus Discussion

1. Orient the class to the purpose and the format of a class consensus discussion. You may say something like this:

"We are going to use a **Class Consensus Discussion**, just like we did a few days ago, to learn about all the thinking in the room. This time we are going to come to some decisions about what explains the existence of water in different phases on different celestial objects in our solar system."

You may decide to walk students through the entire poster again, or take them through the steps as you facilitate it.

Class Consensus Discussion Steps

- 1) we select a few different groups' ideas.
- 2) The first group shares out their work.

Routine



Class Consensus Discussions provide an opportunity for groups to share out around their sensemaking and for other groups to list, summarize, and ask questions after each share.

We recommend you have groups display their data and models while they share their ideas. A discussion with no

- 3) One person repeats or reiterates what the first group shared.
- 4) Class members ask clarifying questions about the work.

Repeat steps 2-4 for each group that is sharing work.

- 5) Everyone confers in table groups.
- 6) Engage in whole-class discussion about the ideas that were shared, in order to come to agreement.

2. Call on the selected students to share their solar system models with the class. This should not be done randomly; it should be based on observations and conferring done as students were working. The point of this discussion is to elevate ideas that move the class towards greater understanding of the role orbits play in stability and change of planet temperature and the potential existence of liquid water. The decision about which models to share with the class should be based on both the ideas circulating in the classroom *and* the goals of this part of the 5E sequence. (See the previous Look and Listen For for suggestions.)
3. Ask the first group to share their model. You can do this by:
 - Projecting using a document camera; OR
 - Copying the models to be shared and passing them out to the class; OR
 - Taking a picture of each model and projecting them as slides.
4. With each group that presents, pause and reflect on the important features of the motions of orbiting objects that explain stability and change of temperature and phase of water on objects in our solar system.
5. Proceed through the steps in the Consensus Discussion Steps. During the whole-class discussion, there will be opportunities to identify important terms and concepts that emerge in the discussion. Sometimes, important points get buried in student talk, so be sure to facilitate the conversation so that key ideas emerge. The following prompts may help with surfacing key ideas through the lens of Stability and Change:
 - What patterns did you see in the orbital data? How did graphing the orbital data help you to

visual component can leave out a number of students.

Integrating Three Dimensions



The prompt about patterns in the Class Consensus Discussion are in support of students' consideration of **CCC #1 - Patterns**. They are asking students to reflect upon how a mathematical representation (graphs) was useful for identifying evidence of patterns. Be sure to assess whether students apply their understanding of the high school patterns element addressed in the How the Sun Works 5E. At this point they should name the specific empirical evidence they used to identify different patterns. If they do not, prompt them to do so.

Access for Multilingual Learners



Rather than assigning a list of vocabulary words—a technique that rarely works for learning new vocabulary—this activity allows language learners to learn vocabulary from context, after having made sense of an experience, which may be particularly helpful for **transitioning** language learners,

- identify evidence of these patterns?
- What determines whether a planet or object's temperature is stable or unstable throughout its entire orbit?
- What factors are causing the phase of water to change or stay the same?

who already have some mastery of language.

Take Time for These Key Points



Pause the discussion and ask for clarification, particularly of the following **key points**. Note that the terms in bold are ones that students probably will figure out but not know the formal vocabulary work. It is appropriate to provide these terms at this point!

- Graphing the data provided a visual that made it much easier to see evidence of the patterns.
- Objects in our solar system orbit around the Sun at different distances and this has an effect on the temperature and phase of water on those objects.
- Most of the objects in our solar system have orbits that are very close to being circular, so they maintain a stable enough temperature for water to stay mostly in one phase throughout an entire revolution around the Sun.
- Some planets in the solar system exist in a **habitable zone**; when looking for exoplanets that can support life we should look for planets in that zone.

6. Have students talk to each other in groups and decide 1-3 things that should be in the class consensus model. Use the Group Learning Routine, **Domino Discover** to surface student ideas. Document these ideas where all students can see them.

Summary

1. Ask students to work independently to complete the Summary Task.
2. Use the answers in the Summary Task to make decisions about which ideas may need to be revisited or explored more in coming parts of this 5E Instructional sequence.

Explore 2

Developing a mathematical model for solar system orbits

Students **use algebraic thinking to examine graphs** from the Explore 1 in order to **create a mathematical representation** that **allows them to identify the exact pattern** in the relationship between **orbital period** and **average distance between a planet/object and its star**.

Preparation

Student Grouping

☐ Pairs

Routines

None

Literacy Strategies

None

Materials

Handouts

- ☐ Exoplanets and their Stars Investigation
- ☐ Making Sense of the Exoplanets and their Stars Investigation
- ☐ Exoplanets and their Stars Investigation Rubric

Lab Supplies

None

Other Resources

- ☐ Computers with Excel or Google spreadsheet
- ☐ [Exoplanet Detection: the Transit Method](#)

Launch

1. Explain that the next part of the investigation will address orbital patterns of exoplanets, and ask students for their thoughts on why we are moving from our solar system to looking at planets orbiting other stars.
2. Tell students that one of the ways exoplanets are detected is called the **transit method**, and while this method allows us to detect exoplanets and make some determinations about them, it does not allow us to collect data about the distance at which the exoplanets orbit from their star. Explain that students

will be making sense of a computational model of the transit method in order to see if it provides data that is useful for determining characteristics of orbits.

Implementation Tip



The term **transit method** is not a key concept or piece of vocabulary for this unit. Rather, it is a shared tool students are going to be using. For that reason, it is fine to teach this as a term without exploration first.

Working with a Computational Model

1. Provide students with the NASA simulations [Exoplanet Detection: the Transit Method](#), [Transit Method Different Planet Sizes](#), and [Transit Method Multiple Planets](#). Have students work in pairs to use the simulation to look at different methods (single planet, different size planets, and multiple planets), with these guiding questions:
 - What patterns do you see in the simulation? What is your evidence?
 - What inferences can we make based on these patterns? Why?
1. As students work, confer with them to ensure they are trying all the possible ways of using the simulation and surfacing all the key points. Refer to the Look & Listen for points below.
2. Project the NASA simulation on the board, and ask several pairs to share their findings. Be strategic in selecting students to share out, so that a range of ideas are surfaced.

Look & Listen For



- When the planet passes in front of its star or how often the planet passes in front of its star. We know this because the light from the star decreases the same amount each time a planet passes in front of it
- The relative size of the planets. We can figure this out by observing how much the light from the star decreases – bigger planets make the light decrease more
- The number of planets orbiting around a star. We know this by watching to see if the light from the star decreases by different amounts

Listen especially for the key information below, which is required for success in the next phase

Integrating Three Dimensions



The prompts about patterns support students' consideration use of **CCC #1 - Patterns**. Students should, at this point, be used to seeking out patterns as a way to make sense of simulations and data sets!

Access for Multilingual Learners



For **emerging language learners**, hearing from classmates describing the activity they just completed and their findings provides comprehensible input that supports their language development. It also gives **transitioning language learners** an opportunity to rehearse and try out ideas in front of the

- The most important idea here is that the transit method allows us to determine how often a planet passes in front of a star.
 - If this idea does not surface, show students the transit method graph below. Ask them to approximate how often the brightness of this star decreases due to Kepler 5b passing in front of it (3.5 days). (This activity is available from NASA JPL at the following link: <https://www.jpl.nasa.gov/edu/teach/activity/exploring-exoplanets-with-kepler/>)
- After there is a shared understanding of this idea, students should be clear that the orbital variable that relates to this is **orbital period**. This is the variable they will work with in the next part of the Explore 2 phase.

class, which is more supportive than extemporaneous talk or cold calling.

Analyzing Data about Orbital Motion

1. Provide students with some framing for analyzing data about orbital motion. As a way of narrating this transition, show some of the artifacts (graphs, verbal explanations, etc.) of ideas that have been figured out so far, and narrate the next step: "The transit method does not tell us anything about the distance at which exoplanets orbit around their star. We just figured out that the transit method shows there is something orbiting, but that's all! Remember in Explore 1 we observed a pattern in the relationship between average distance at which a celestial object orbits around the Sun and its orbital period? Let's go back and think about that relationship some more."
2. Display the graph and ask students to turn and talk to a partner about how they can use the evidence of a pattern in the graph to make claims about the average distance at which EXOPLANETS orbit from their respective stars.
3. Ask a few pairs to share out and listen for them to say that they can see at which average distance the exoplanet's orbital period intersects the graph. Acknowledge the idea and suggest that the class tries this method.
4. Ensure that all students have the graph accessible and ask them to discuss the following prompt in pairs:
 - If an exoplanet's orbital period is 50 Earth years, what is the avg distance at which it orbits its star?

Integrating Three Dimensions



This is an opportunity to support students' use of **CCC #1 - Patterns** alongside **SEP #2 - Developing & Using Models**. Students have had multiple opportunities so far in this unit to seek out patterns as a way to make sense of simulations and data sets. Deriving a mathematical equation from data will support students in understanding the role mathematical representations play in allowing them to identify patterns that otherwise could not be identified. In this case allowing them to describe the pattern in the relationship

5. Have several pairs share out and listen for a range of responses between 12.5 and 14 AUs. Ask students why there were a range of responses and why this might be a problem when predicting the average distance at which an exoplanet orbits from its host star. Listen for students to say that the graph makes it challenging to decide at exactly which average distance the orbital period intersects the graph and that would lead to less confidence in their claims.
6. Explain that a way scientists address this challenge is by representing patterns through equations we can derive from the data.
7. Have students revisit their spreadsheets [Excel Spreadsheet for Planet Data Analysis Graphing](#) (using Excel, Google Sheets, or another system) and derive an equation that describes the relationship between the average distance at which a celestial object orbits the Sun and its orbital period.
8. As groups are working, support them with figuring out the mathematical model equation. It may be that some pairs need more one-on-one support with using the spreadsheet to create a model. See the Tip box below with guidance on this.
9. Ensure that all groups have come up with an equation that represents the data.

between orbital period and average distance more accurately. They will explicitly reflect upon this in the Explain 2 phase.

Implementation Tip



This is their first time deriving an equation as a mathematical representation. This activity, therefore, is more guided than subsequent ones may be. The goal here is to move students toward understanding the necessity of a mathematical representation (equation) for identifying some patterns in data by the end of the Explain 2 phase. At this point, they don't necessarily need to internalize the process for deriving the equation or understand the function that best fits the graph.

Implementation Tip



Since this may be students' first time using a spreadsheet to derive an equation from a data, consider facilitating this as a pair activity, with whole-class check-ins to make sure everyone is following the process.

The following are steps for deriving equations from data using Google Sheets:

Graph the data (average distance from the Sun vs period of revolution)	
Double click on the graph and select 'Series' from the menu on the right	
Click 'Trendline' on the bottom of the 'Series' menu	
Try the different types of trendlines until you find one that best fits the graph. 'Power Series' should be an exact fit.	
Click on the drop down for 'Label' and select 'Use Equation'. Then click on 'Show R ² '.	
The equation and R² value below should appear at the top of the graph. X = orbital period X ^{1/3} (cube root) = the average distance between the celestial object and its star, or semi-major axis. Note: In this image R does not represent average distance. An R² value of 1 indicates the trendline line chosen is a perfect fit for the graph.	

10. Provide students with the handout *Making Sense of the Exoplanets and their Stars Investigation*. Have students work individually to complete the see-think-wonder.
11. Make a determination about whether to have groups share out or just move on. The share may be redundant if all groups are confident in their mathematical equation, and the teacher has determined the mathematical equation to be correct.
12. Provide students with *Exoplanets and their Stars Investigation Rubric*. Ask students to use the investigation rubric to self and peer assess their progress on engaging with the investigation individually and as a group.

Explain 2

Using a mathematical representation to determine the orbit of exoplanets

Students **use a mathematical equation** representing the **relationship between orbital semi major axis and orbital period (Kepler's Third Law)** to **make predictions** about exoplanets' **orbital semi major axis** based on each exoplanet's **orbital period**.

Preparation

Student Grouping

☐ Pairs

Routines

☐ Class Consensus Model

Literacy Strategies

None

Materials

Handouts

- ☐ Kepler's Third Law
- ☐ Applying our mathematical model of orbits to exoplanets
- ☐ Summary Task

Lab Supplies

None

Other Resources

- ☐ Kepler's Third Law Calculator
- ☐ Which Exoplanets Stay Within the Habitable Zone Example Class Response

Launch

1. Frame the next activity for students: "Now that we have established a mathematical representation of the relationship between a planet's average distance from its host star and orbital period, how can we use that to make a claim about objects orbiting *other* stars?" Have students share their thinking about applying what we figured out to other stars in the universe.
2. Share with students that an astronomer named Johannes Kepler went through a similar process to the one they just did to derive mathematical equations like the ones they did. The whole point of coming up with these equations is being able to make predictions about other situations, where we don't have as

much information.

3. Have students review the law and write the equation as a verbal explanation on the *Kepler's Third Law* handout. Ensure that students see the connection before moving on.
4. Point out that Kepler's equation had one other variable, the mass of the Sun. Ask students how we might account for stars with different masses. Explain that the mass of other stars will be provided in solar masses.

Applying Mathematical Models to Exoplanets

1. Have students work in pairs to complete the *Applying our mathematical model of orbits to exoplanets* handout.
2. As students are working, consider which groups should share in the class consensus discussion in order to surface many viewpoints.

Look & Listen For



As students are completing their responses, look and listen for these ideas in response to the questions below, which will be important to surface in the Class Consensus Discussion:

1. The graph of orbital period vs semi major axis length allowed us to identify a pattern in the relationship between these two variables. How did our mathematical model (Kepler's Third Law) help us describe this pattern more accurately than the graph?

The graph only lets us see that as the average distance at which a planet orbits from its star increases, the orbital period increases too. The mathematical model let us see exactly how those variables relate to each other by including exponents.

2. Based on your calculations, which exoplanets travel within the habitable zone when they are at the average distance from its star? Be sure to cite evidence for your claim.

Integrating Three Dimensions



The prompts in the handout are designed to get students thinking about and using **CCC #1 - Patterns and CCC # 3 Scale, Proportion, and Quantity** to make sense of the analysis of these planets. The mathematical model that describes the pattern in the relationship between orbital period and average distance of celestial objects, can be used to predict the average distance at which exoplanets orbit around their stars which is an example of using algebraic thinking to examine scientific data and predict the effect of a change in one variable on another, an important element of **CCC # 3 Scale, Proportion, and Quantity**.

Kepler 186 f - when you plug its orbital period of 0.3556 years and its host star mass of 0.544 Sun masses into Kepler's 3rd law, I calculate that the average distance at which it orbits its host star is 0.4097 AU, which falls within the habitable zone of 0.242 - 0.464 AU.

Kepler 442 b - when you plug its orbital period of 0.3078 years and its host star mass of 0.61 Sun masses into Kepler's 3rd law, I calculate that the average distance at which it orbits its host star is 0.3866 AU, which falls within the habitable zone of 0.252 - 0.531 AU.

TOI-2257 b - when you plug its orbital period of 0.0964 years and its host star mass of 0.33 Sun masses into Kepler's 3rd law, I calculate that the average distance at which it orbits its host star is 0.1453 AU, which falls within the habitable zone of 0.079 - 0.167 AU.

Trappist 1f - when you plug its orbital period of 0.0252 years and its host star mass of 0.089 Sun masses into Kepler's 3rd law, I calculate that the average distance at which it orbits its host star is 0.0384 AU, which falls within the habitable zone of 0.023-0.048 AU.

Trappist 1g - when you plug its orbital period of 0.034 years and its host star mass of 0.089 Sun masses into Kepler's 3rd law, I calculate that the average distance at which it orbits its host star is 0.0469 AU, which falls within the habitable zone of 0.023-0.048 AU.

3. Does this mean that they orbit within the habitable zone throughout their orbital period of revolution around their host star? What other information would you like to have?

Not necessarily. Their orbits could be like Comet Borrelly. we still need to figure out their orbital shapes!

Class Consensus Discussion

1. Orient the class to the purpose and the format of a class consensus discussion. You may say something like this:

"We are going to use a **Class Consensus Discussion**, just like we did a few days ago, to learn about all the thinking in the room and come to some decisions about what our mathematical model has helped us figure out about the orbit, and the potential for water, on exoplanets in our data set." The share will be about responses to the following prompts from the *Explain 2: Applying our mathematical model of orbits to exoplanets* handout:

- The graph of orbital period vs semi major axis length allowed us to identify a pattern in the relationship between these two variables. How did our mathematical model (Kepler's Third Law) help us describe this pattern more accurately than the graph?
- Based on your calculations, which exoplanets travel within the habitable zone when they are at the average distance from their star? Be sure to cite evidence.
- Does this mean that they orbit within the habitable zone throughout their orbital period of revolution around their star? What other information would you like to have?

You may decide to walk students through the entire poster again, or take them through the steps as you facilitate it.

2. Select two or three groups' to share their responses with the class. At this point, do not select them randomly. The point of this discussion is to elevate ideas that move the class towards greater understanding of how a mathematical model can be used to generate evidence to make a claim about stability and change of planet temperature and the potential existence of liquid water on planets in the exoplanet data set. The decision about which groups share with the class should be based on both the ideas circulating in the classroom *and* the goals of this part of the 5E sequence.
3. Ask the first group to share their responses. You can do this by:
 - Projecting using a document camera; OR
 - Copying their responses to be shared and passing them out to the class; OR
 - Taking a picture of responses and projecting them as slides.
4. With each group that presents, pause and reflect on the important features of the motions of orbiting objects that explain stability and change of temperature and phase of water on objects in our solar system.
5. Proceed through the steps in the Consensus Discussion Steps. During the whole-class discussion, there will be opportunities to identify important terms and concepts that emerge in the discussion.

Sometimes, important points get buried in student talk, so be sure to facilitate the conversation so that key ideas emerge.

Implementation Tip



Class Consensus Discussions provide an opportunity for groups to share out around their sensemaking and for other groups to list, summarize, and ask questions after each share.

We recommend you have groups display their data and responses while they share their ideas. A discussion with no visual component can leave out a number of students.

which exoplanets stay within the habitable zone throughout their orbit around their stars?

Kepler 186 - f

Kepler 442 - b

TOI-2257 b

Trappist 1 - f

Trappist 1 - g

All are within the habitable zone of their solar system when they are orbiting at their average distance from their star.

we don't know for sure if they stay within the habitable zone throughout an entire orbit because we don't know the max and min distance at which they orbit (we don't know the orbit shape)

we need to know the shape of their orbits or the min and max distance

Take Time for These Key Points



Pause the discussion and ask for clarification, particularly of the following **key points**:

- While the graph of orbital period vs semi major axis length allowed us to identify a pattern in the relationship between these two variables, the mathematical model described this pattern more accurately because it included exponents and a coefficient of star mass.
- By plugging in the orbital period for each planet into our mathematical model, we were able to generate additional data, average distance from its star, for each planet.
- This data served as evidence to make a claim about whether each planet maintains a stable temperature that would allow liquid water to exist.
- Kepler 186 - f, Kepler 442 - b, TOI-2257 b, Trappist 1 - f, and Trappist 1 - g are all orbiting within the habitable zone when they are at their average distance from the Sun.
- We don't know for sure if they stay within the habitable zone throughout an entire orbit because we don't know the max and min distance at which they orbit (we don't know the orbit shape).
- We need to know the shape of their orbits or the min and max distance.

Revisiting What Counts as an Evidence-Based Claim

1. Ask students to share their current thinking about what we are trying to figure out (*Does the exoplanet have stable temperature and maintain liquid water?*).
2. Facilitate a class discussion around what constitutes an evidence-based claim using their common experiences during this most recent 5E as a reference.

What Counts as an Evidence-Based Claim?

You found information from a book or a reliable source.

The evidence comes from an experiment or investigation you did.

The claim is not just someone's opinion.

Many scientists can agree on that interpretation.

Patterns in data count as evidence you can use for claims.

Patterns in data can count as evidence for a claim. But you have to have evidence for the pattern too.

Evidence for a claim comes from other scientists' data.

Evidence for a claim comes from patterns observed in models based on data.

A model based on evidence, like a diagram that shows how something works.

We can use technology like a spreadsheet to derive mathematical models that allow us to generate data.

Data generated from a mathematical model can serve as evidence for making claims or predictions.

Sometimes you need two pieces of data together in order for either to serve as evidence.

Summary

1. Ask students to work independently to complete the Summary Task.
2. Use the answers in the Summary Task to make decisions about which ideas may need to be revisited or explored more in upcoming parts of this lesson.

Elaborate

Constructing orbits of exoplanets to determine whether liquid water can exist on each planet

Students **use a mathematical model** (**Kepler's First Law**) to **describe the relationship** between **orbital eccentricity, major axis, and distance between foci** of a **planet/object revolving around its host star**. Students then **use the mathematical model** and **algebraic thinking** to construct exoplanet orbits, allowing them to **make predictions** about which exoplanets stay within the habitable zone of their stars throughout an entire orbit.

Preparation

Student Grouping

☐ Pairs

Routines

☐ Think-Talk-Open Exchange

Literacy Strategies

☐ Text Annotation

Materials

Handouts

- ☐ How do we determine shapes of orbits?
- ☐ Which exoplanets orbit in the habitable zone?
- ☐ Orbital Data Optional Extension

Lab Supplies

None

Other Resources

☐ [Classroom Demonstration: Elliptical Orbits](#)

Launch

1. Frame for students where we are in figuring things out: Which exoplanets in the performance task data set stay within the habitable zone throughout an entire orbit? Tell them that the only exoplanet orbital data they have access to is the orbital period and the average distance at which each exoplanet orbits around its star, which they calculated using the mathematical model they developed.
2. Ask students to think about what they have done so far in this 5E investigation and brainstorm ideas for how we can figure out whether the exoplanets stay within the habitable zone throughout an orbit.

Look & Listen For



While pairs are sharing their responses, listen for these ideas, to provide a bridge to the Elaborate phase:

- We can find a relationship between the variables we know and the variables we are trying to figure out so we can develop a mathematical model
- We can use the mathematical model to calculate the variables we are trying to figure out and maybe use this data as evidence for a claim

3. Leverage the ideas above to transition to the next part of this phase, focused on Kepler's First Law.

Reading about Kepler's Law

1. Tell students that Johannes Kepler also developed a mathematical model that describes the relationship between the distance at which a planet orbits, the mass of objects in the solar system, and the shape of the planet's orbit. "While we have been able to determine which exoplanets in our data set are within the habitable zone when they are at their average distance from their host stars, we still need to determine whether they stay within the habitable zone throughout an entire revolution. Thankfully, Johannes Kepler also developed a mathematical model to describe and calculate orbital shapes."
2. Have students read the text about Kepler's First Law of Planetary Motion, using a **text annotation** strategy to identify new ideas and ideas that are relevant to what we are trying to figure out.

Differentiation Point



These two pages of the student materials include particularly dense text and introduce several tier three vocabulary terms. Consider providing small group instruction to **multilingual learners** and **below grade-level readers**. You can support them in navigating the text, by conducting a read aloud and prompting them to make connections between the written text being read aloud, diagrams, and the algebraic equation.

Constructing an Orbit

1. Put students into groups of three to work on calculating the distance between foci in each exoplanet's solar system and construct exoplanet orbits. Using Earth as an example, demonstrate how to construct an orbit. For teacher guidance around constructing orbits, watch the Classroom Demonstrations - Elliptical Orbits video. Have students follow along for practice while you:
 - Calculate the distance between foci based on the length of Earth's major axis and eccentricity
 - Use the distance between foci and the length of the major axis (length of string loop) to construct Earth's orbit around the Sun
 - Represent the inner and outer limits of the habitable zone and determine whether Earth stays within the habitable zone throughout its orbit around the Sun

Implementation Tip



This activity may feel a bit more teacher-led than other parts of this unit. Keep in mind that teaching students how to figure out orbital shape based on data is not an end in itself! Rather, this is another opportunity to interpret a mathematical model and use it as a tool to make evidence-based claims.

Routine



Think-Talk-Open Exchange is a structured discussion routine that allows students the opportunity to share with others and gain feedback on their ideas by finding similarities and differences, piecing together disparate bits of information, or reconciling different interpretations. Refer to the Earth & Space Science Course Guide for support with this routine.

2. Point out to students that Earth does fall within the habitable zone throughout its orbit because the low eccentricity means it has a low distance between foci, and Earth's distance throughout orbit therefore does not change a lot.
3. Ask students to name which of the other exoplanets have a lower eccentricity than Earth's and therefore will also not change distance very much and stay within the habitable zone. Listen for students to say that trappist 1-f and trappist 1-g both have eccentricities lower than Earth. Let them know that for that reason they don't have to construct the orbit and can assume those planets stay within the habitable zone.
4. Have each student construct the orbit for Kepler 186 -f, Kepler 442-b, and TOI-2257.
5. After students have completed all three orbits, have groups of three use the group learning routine **Think-Talk-Open Exchange** to share what they figured out:
 - how they constructed the orbit for their assigned exoplanet;
 - how each variable in Kepler's First Law of Planetary Motion is represented in their model; and
 - their conclusions about whether the planet stays within the habitable zone throughout its orbit.
6. Use the table below for guidance on the answers to expect from students.

Take Time for These Key Points



Pause the discussion and ask for clarification, particularly of the following **key points**:

- We algebraically manipulated Kepler's First Law to isolate the unknown variable f , then plugged in our known values, e and d , in order to calculate f .
- The distance between the two centers of mass in the solar system, f , is represented by the two thumbtacks.
- The length of the orbit's major axis, d , is represented by the longest distance between two points on the orbit.
- The eccentricity of the orbit, e , is represented by how circular or elliptical the orbit is.
- Kepler 186 -f, Kepler 442 b, Trappist - 1f, and Trappist - 1g have a very circular orbit or low orbital eccentricity, so they stay within the habitable zone of their solar system throughout an entire orbit of their host star. TOI-2257 b's orbital shape is very oval like (eccentricity is very high), so while it does orbit within the habitable zone during part of its orbit, it does not stay within the habitable zone of its solar system throughout an entire orbit around its host star.

which exoplanets stay within the habitable zone throughout their orbit around their stars?

Kepler 186 -f

Kepler 442 b

Trappist - 1f

Trappist 1 - g

All have a very circular orbit or low orbital eccentricity, so they stay within the habitable zone of their solar system throughout an entire orbit of their host star.

Orbital Data

Orbital eccentricity, length of major axis, and the distance between foci data for the exoplanets below comes from the NASA exoplanet archive. The length of string and distance between foci for student creation of orbits was calculated by multiplying all actual values for each planet by the same factor. Values to be used for student activity are in columns highlighted with blue.

Note: Habitable Zone should be measured from the focus that represents the star.

Celestial Object	Major Axis (million km)	Scaled Length of Major Axis (cm)	Length of String (cm)	Eccentricity of Orbit	Distance Between Foci (million km)	Scaled Distance Between Foci (cm)	Habitable Zone Range (million km)	Scaled Habitable Zone Range (cm)
Earth	299.2	60	30.5	0.0167	4.99664	1	142.12 - 172.04	28.5 - 34.5
Kepler 186 f	122.7	50	26	0.04	4.91	2	36.20 - 69.41	14.8 - 28.3
Kepler 442 b	137.330	50	26	0.04	4.63	2	37.70 - 79.44	16.3 - 34.4
TOI-2257	43.47	25	18.7	0.4960	21.56	12.4	11.82 - 24.98	6.8 - 14.4
Trappist 1 f	11.49	5.7	2.9	0.010	.114	.06	3.44 - 7.18	1.7 - 3.6
Trappist 1 g	14.03	7.0	3.5	0.002	.028	.01	3.44 - 7.18	1.7 - 3.6

Differentiation Point



Extension Activity

If students complete the Elaborate phase early and demonstrate mastery, remind them that one of the questions the class had about exoplanets they are investigating is whether they have an atmosphere. Provide data for planets in our solar system, including diameter, mass, and density data of different planets. Challenge them to look for patterns in our solar system data associated with the existence of an atmosphere on planets in our solar system. They should see a pattern in the relationship between mass and existence of an atmosphere and diameter and the existence of an atmosphere. Encourage them to think about why these relationships exist and use those patterns to make claims about the existence of an atmosphere on exoplanets in the performance task data set.

Orbital Data Optional Extension

1. For New York Teachers, consider using the *Orbital Data Optional Extension* to apply their learning to this table from the New York State Earth & Space Science Reference Tables.

Evaluate

Developing or revising models to include new ideas about orbits

Students **develop new models or revise existing ones** to represent the idea that **stable** temperatures and the maintenance of liquid water result from **features of the motions of orbiting objects**.

Preparation

Student Grouping

- ☐ Small groups of 3-4 students (same groups from Unit Launch)

Routines

- ☐ Idea Carousel
- ☐ Domino Discover

Literacy Strategies

None

Materials

Handouts

- ☐ Planets and Orbits Model Rubric
- ☐ Connect to the Performance Task: Planets and Orbits

Lab Supplies

None

Other Resources

- ☐ Driving Question Board
- ☐ Groups' revised models (charts)
- ☐ Class wide evidence-based claim poster
- ☐ *Connect to the Performance Task: Planets and Orbits Student Work*

Revise and Share Models

1. Ask students to consider their responses to these questions and what they have figured out about orbits, then represent their ideas about why the Sun has been able to support a planet where life has been able to exist and evolve on their initial group models from the performance task launch. These should go onto new pieces of chart paper
2. Facilitate student critique of one another's models through the Group Learning Routine **Idea Carousel**.

Routine

Have students annotate other groups' models using post-its. Each post it should have a symbol and comment from each of the following categories:

- a. ✓ Write a check on sticky notes with comments about ideas represented in the model that resonate.
- b. + Write a plus symbol on sticky notes with comments about ideas that should be added to the model.
- c. ? Write a question mark on sticky notes with comments about ideas that you don't think are relevant to the model.
- d. Δ Write a delta symbol on sticky notes with comments about suggestions for how to clarify an idea or represent it more clearly.

Look & Listen For



While students are engaged in the Idea Carousel, listen for the following ideas. Where needed, discuss with groups what is coming up in their models, to ensure these points emerge in the classroom.

- The habitable zone range in our solar system is 142.12-172.04 million km and Earth orbits the Sun at an average distance of 150 million km, which is within the habitable zone.
- Earth's orbit is very circular, with an eccentricity of 0.0167, so it stays within the habitable zone throughout an orbit and maintains mostly liquid water. This is one of the major reasons earth can sustain life.



This routine allows for each group to give and receive warm feedback and suggestions around content and clarity in their model, as well as pose and respond to questions. The repeated use of **Idea Carousel** should support students in gaining familiarity with the routine and getting better at using it as a tool to surface thinking.

Access for Multilingual Learners



The routine **Idea Carousel** is ideal for **emerging language learners**. Students with only receptive language can simply engage by listening and adding annotations

Routine



The **Domino Discover** group learning routine is an opportunity to surface students' thinking to the whole class and the teacher. It allows students to learn from each other and for the teacher to assess whether the class is ready to move to the next phase of instruction. Refer to the Earth & Space Science Course Guide for support with this routine.

Return to the Performance Task

1. Using what they learned in the Idea Carousel, have students independently refine their models in their performance task research organizer. See the sample response below for an idea of the types of responses to expect at this stage in the unit.
2. After students complete their work, support them to use the rubric for this learning sequence as a self-reflection tool. Here are some ways the rubric can be used:
 - a. Have students complete a self-assessment using the rubric.
 - b. Create a piece of student work that is a fictional composite of a few different students' work, and complete a critique as a class, while students critique their own work using the rubric.
 - c. Collect all the work and score the work, with an eye to how accurate students' self-assessments seem to be.
3. Assign students to partnerships and have them review one another's work and self-assessment and provide feedback on the accuracy of the self-assessment.
4. Give each student an opportunity to revise their model using what surfaced from their self-assessment and/or feedback.
5. Let students know that they will now use all the ideas represented in their models as a lens to analyze and interpret exoplanet data and finalize their performance task argument.

Access for All Learners



All students have learned and probably been curious about planets and the possibility of life on other planets. They've all certainly experienced and enjoyed water throughout their lives. The prompts at the end of the rubric are designed to support students in pausing to reflect about why their new ideas are relevant to their lives. After students complete the reflection prompts independently, consider facilitating a whole class share around why learning during this 5E is relevant to students' lives and displaying their responses somewhere in the classroom. This can help keep students motivated to continue the unit investigation.

Standards in Planets and Orbits 5E

Performance Expectations

HS-ESS1-4

Use mathematical or computational representations to predict the motion of orbiting objects in the solar system.

Clarification Statement: Emphasis is on Newtonian gravitational laws governing orbital motions, which apply to human-made satellites as well as planets and moons.

Assessment Boundary: Mathematical representations for the gravitational attraction of bodies and Kepler's Laws of orbital motions should not deal with more than two bodies, nor involve calculus.

Aspects of Three-Dimensional Learning

Science and Engineering Practices

Disciplinary Core Ideas

Crosscutting Concepts

Developing and Using Models

- Develop, revise, and/or use a model based on evidence to illustrate and/or predict the relationships between systems or between components of a system. SEP2(3)

Using Mathematics and Computational Thinking

- Use mathematical, computational, and/or algorithmic representations of phenomena or design solutions to describe and/or support claims and/or explanations. SEP5(2)

ESS1.B Earth and the Solar System

- Kepler's laws describe common features of the motions of orbiting objects, including their elliptical paths around the Sun. Orbits may change due to the gravitational effects from, or collisions with, other objects in the solar system. ESS1.B(1)

Patterns

- Mathematical representations are needed to identify some patterns. CCC1(4)

Scale, Proportion, and Quantity

- Algebraic thinking is used to examine scientific data and predict the effect of a change in one variable on another (e.g., linear growth vs. exponential growth). CCC3(5)
-

Assessment Matrix

	Engage	Explore/Explain 1	Explore/Explain 2	Elaborate	Evaluate
Developing and Using Models	Initial Model	<i>Revising Solar System Models</i>	<i>Applying our mathematical model of orbits to exoplanets</i>	<i>Which exoplanets orbit in the habitable zone?</i>	<i>Connect to the Performance Task: Planets and Orbits</i>
Using Mathematics and Computational Thinking		<i>Orbital Data Investigation Making Sense of the Orbital Data Investigation Revising Solar System Models</i>	<i>Exoplanets and their Stars Investigation Making Sense of the Exoplanets and their Stars Investigation Applying our mathematical model of orbits to exoplanets</i>	<i>Which exoplanets orbit in the habitable zone?</i>	
ESS1.B Earth and the Solar System	Initial Model	<i>Orbital Data Investigation Making Sense of the Orbital Data Investigation Revising Solar System Models Summary Task</i>	<i>Exoplanets and their Stars Investigation Making Sense of the Exoplanets and their Stars Investigation Applying our mathematical model of orbits to exoplanets</i>	<i>Which exoplanets orbit in the habitable zone?</i>	<i>Connect to the Performance Task: Planets and Orbits</i>
Patterns		<i>Making Sense of the Orbital Data Investigation Revising Solar System Models Summary Task</i>	<i>Exoplanets and their Stars Investigation Summary Task</i>		<i>Connect to the Performance Task: Planets and Orbits</i>
Scale, Proportion, and Quantity		<i>Orbital Data Investigation Making Sense of the Orbital Data Investigation</i>	<i>Exoplanets and their Stars Investigation Making Sense of the Exoplanets and their Stars Investigation Applying our mathematical model of orbits to exoplanets Summary Task</i>	<i>Which exoplanets orbit in the habitable zone?</i>	<i>Connect to the Performance Task: Planets and Orbits</i>

Common Core State Standards Connections

	Engage	Explore/Explain 1	Explore/Explain 2	Elaborate	Evaluate
Mathematics		MP2 MP4	MP2 MP4 HSA-SSE.A.1 HSA-CED.A.2	MP2 MP4 HSA-SSE.A.1 HSA-CED.A.4	MP2 MP4
ELA/Literacy		RST.9-10.1 RST.9-10.7 WHST.9-10.9 SL.9-10.4	RST.9-10.7 WHST.9-10.1 WHST.9-10.9 SL.9-10.5	RST.9-10.1 WHST.9-10.9	WHST.9-10.9 SL.9-10.5